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21 Author

BA Johnson.

22 Date of Revision

26 February 2009.

Acetic Acid, Glacial

1 Nonproprietary Names

BP: Glacial Acetic Acid

JP: Glacial Acetic Acid

PhEur: Acetic Acid, Glacial

USP: Glacial Acetic Acid

2 Synonyms

Acidum aceticum glaciale; E260; ethanoic acid; ethylic acid; methane carboxylic acid; vinegar acid.

See also Sections 17 and 18.

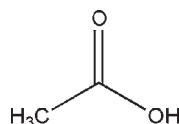
3 Chemical Name and CAS Registry Number

Ethanoic acid [64-19-7]

4 Empirical Formula and Molecular Weight

C₂H₄O₂ 60.05

5 Structural Formula



6 Functional Category

Acidifying agent.

7 Applications in Pharmaceutical Formulations or Technology

Glacial and diluted acetic acid solutions are widely used as acidifying agents in a variety of pharmaceutical formulations and food preparations. Acetic acid is used in pharmaceutical products as a buffer system when combined with an acetate salt such as sodium

acetate. Acetic acid is also claimed to have some antibacterial and antifungal properties.

8 Description

Glacial acetic acid occurs as a crystalline mass or a clear, colorless volatile solution with a pungent odor.

9 Pharmacopeial Specifications

See Table I.

Table I: Pharmacopeial specifications for glacial acetic acid.

Test	JP XV	PhEur 6.0	USP 32
Identification	+	+	+
Characters	+	+	—
Freezing point	≥ 14.5°C	≥ 14.8°C	≥ 15.6°C
Nonvolatile matter	≤ 1.0 mg	≤ 0.01%	≤ 1.0 mg
Sulfate	+	+	+
Chloride	+	+	+
Heavy metals	≤ 10 ppm	≤ 5 ppm	≤ 5 ppm
Iron	—	≤ 5 ppm	—
Readily oxidizable impurities	+	+	+
Assay	≥ 99.0%	99.5–100.5%	99.5–100.5%

10 Typical Properties

Acidity/alkalinity

pH = 2.4 (1 M aqueous solution);

pH = 2.9 (0.1 M aqueous solution);

pH = 3.4 (0.01 M aqueous solution).

Boiling point 118°C

Dissociation constant pK_a = 4.76

Flash point 39°C (closed cup); 57°C (open cup).

Melting point 17°C

Refractive index n_D²⁰ = 1.3718

Solubility Miscible with ethanol, ether, glycerin, water, and other fixed and volatile oils.

Specific gravity 1.045

11 Stability and Storage Conditions

Acetic acid should be stored in an airtight container in a cool, dry place.

12 Incompatibilities

Acetic acid reacts with alkaline substances.

13 Method of Manufacture

Acetic acid is usually made by one of three routes: acetaldehyde oxidation, involving direct air or oxygen oxidation of liquid acetaldehyde in the presence of manganese acetate, cobalt acetate, or copper acetate; liquid-phase oxidation of butane or naphtha; methanol carbonylation using a variety of techniques.

14 Safety

Acetic acid is widely used in pharmaceutical applications primarily to adjust the pH of formulations and is thus generally regarded as relatively nontoxic and nonirritant. However, glacial acetic acid or solutions containing over 50% w/w acetic acid in water or organic solvents are considered corrosive and can cause damage to skin, eyes, nose, and mouth. If swallowed glacial acetic acid causes severe gastric irritation similar to that caused by hydrochloric acid.⁽¹⁾

Dilute acetic acid solutions containing up to 10% w/w of acetic acid have been used topically following jellyfish stings.⁽²⁾ Dilute acetic acid solutions containing up to 5% w/w of acetic acid have also been applied topically to treat wounds and burns infected with *Pseudomonas aeruginosa*.⁽³⁾

The lowest lethal oral dose of glacial acetic acid in humans is reported to be 1470 µg/kg.⁽⁴⁾ The lowest lethal concentration on inhalation in humans is reported to be 816 ppm.⁽⁴⁾ Humans, are, however, estimated to consume approximately 1 g/day of acetic acid from the diet.

LD₅₀ (mouse, IV): 0.525 g/kg⁽⁴⁾

LD₅₀ (rabbit, skin): 1.06 g/kg

LD₅₀ (rat, oral): 3.31 g/kg

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Acetic acid, particularly glacial acetic acid, can cause burns on contact with the skin, eyes, and mucous membranes. Splashes should be washed with copious quantities of water. Protective clothing, gloves, and eye protection are recommended.

16 Regulatory Status

GRAS listed. Accepted as a food additive in Europe. Included in the FDA Inactive Ingredients Database (injections, nasal, ophthalmic,

and oral preparations). Included in parenteral and nonparenteral preparations licensed in the UK.

17 Related Substances

Acetic acid; artificial vinegar; dilute acetic acid.

Acetic acid

Comments A diluted solution of glacial acetic acid containing 30–37% w/w of acetic acid. See Section 18.

Artificial vinegar

Comments A solution containing 4% w/w of acetic acid.

Dilute acetic acid

Comments A weak solution of acetic acid which may contain between 6–10% w/w of acetic acid. See Section 18.

18 Comments

In addition to glacial acetic acid, many pharmacopeias contain monographs for diluted acetic acid solutions of various strengths. For example, the USP32–NF27 has a monograph for acetic acid, which is defined as an acetic acid solution containing 36.0–37.0% w/w of acetic acid. Similarly, the BP 2009 contains separate monographs for glacial acetic acid, acetic acid (33%), and acetic acid (6%). Acetic acid (33%) BP 2009 contains 32.5–33.5% w/w of acetic acid. Acetic acid (6%) BP 2009 contains 5.7–6.3% w/w of acetic acid. The JP XV also contains a monograph for acetic acid that specifies that it contains 30.0–32.0% w/w of acetic acid.

A specification for glacial acetic acid is contained in the Food Chemicals Codex (FCC).⁽⁵⁾

The EINECS number for acetic acid is 200-580-7. The PubChem Compound ID (CID) for glacial acetic acid is 176.

19 Specific References

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- 4 Lewis RJ, ed. *Sax's Dangerous Properties of Industrial Materials*, 11th edn. New York: Wiley, 2004; 15–16.
- 5 *Food Chemicals Codex*, 6th edn. Bethesda, MD: United States Pharmacopeia, 2008; 12.

20 General References

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21 Author

WG Chambliss.

22 Date of Revision

23 March 2009.

1 Nonproprietary Names

BP: Acetone
PhEur: Acetone
USP-NF: Acetone

2 Synonyms

Acetonom; dimethylformaldehyde; dimethyl ketone; β -ketopropane; pyroacetic ether.

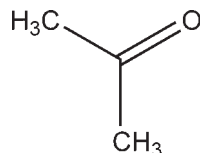
3 Chemical Name and CAS Registry Number

2-Propanone [67-64-1]

4 Empirical Formula and Molecular Weight

C₃H₆O 58.08

5 Structural Formula



6 Functional Category

Solvent.

7 Applications in Pharmaceutical Formulation or Technology

Acetone is used as a solvent or cosolvent in topical preparations, and as an aid in wet granulation.^(1,2) It has also been used when formulating tablets with water-sensitive active ingredients, or to solvate poorly water-soluble binders in a wet granulation process. Acetone has also been used in the formulation of microspheres to enhance drug release.⁽³⁾ Owing to its low boiling point, acetone has been used to extract thermolabile substances from crude drugs.⁽⁴⁾

8 Description

Acetone is a colorless volatile, flammable, transparent liquid, with a sweetish odor and pungent sweetish taste.

9 Pharmacopeial Specifications

See Table I. See also Section 18.

Table I: Pharmacopeial specifications for acetone.

Test	PhEur 6.0	USP32-NF27
Identification	+	+
Characters	+	—
Appearance of solution	+	—
Acidity or alkalinity	+	—
Relative density	0.790–0.793	≤0.789
Related substances	+	—
Matter insoluble in water	+	—
Reducing substances	+	+
Residue on evaporation	≤50 ppm	≤0.004%
Water	≤3 g/L	+
Assay	—	≥99.0%

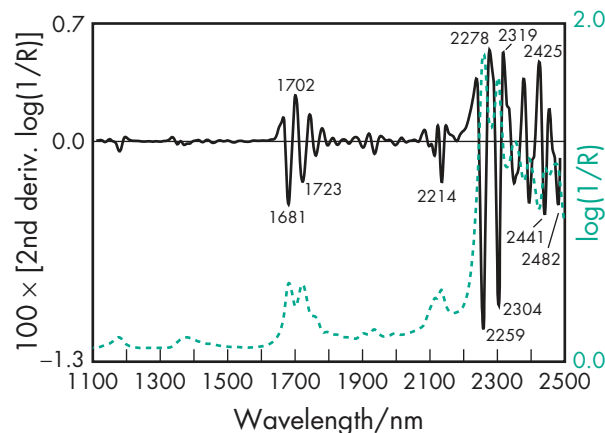


Figure 1: Near-infrared spectrum of acetone measured by transfectance (1 mm path-length).

10 Typical Properties

Boiling point 56.2°C

Flash point -20°C

Melting point 94.3°C

NIR spectra see Figure 1.

Refractive index $n_D^{20} = 1.359$

Solubility Soluble in water; freely soluble in ethanol (95%).

Vapor pressure 185 mmHg at 20°C

11 Stability and Storage Conditions

Acetone should be stored in a cool, dry, well-ventilated place out of direct sunlight.

12 Incompatibilities

Acetone reacts violently with oxidizing agents, chlorinated solvents, and alkali mixtures. It reacts vigorously with sulfur dichloride, potassium *t*-butoxide, and hexachloromelamine. Acetone should not be used as a solvent for iodine, as it forms a volatile compound that is extremely irritating to the eyes.⁽⁴⁾

13 Method of Manufacture

Acetone is obtained by fermentation as a by-product of *n*-butyl alcohol manufacture, or by chemical synthesis from isopropyl alcohol; from cumene as a by-product in phenol manufacture; or from propane as a by-product of oxidation-cracking.

14 Safety

Acetone is considered moderately toxic, and is a skin irritant and severe eye irritant. Skin irritation has been reported due to its defatting action, and prolonged inhalation may result in headaches. Inhalation of acetone can produce systemic effects such as conjunctival irritation, respiratory system effects, nausea, and vomiting.⁽⁵⁾

LD₅₀ (mouse, oral): 3.0 g/kg⁽⁵⁾

LD₅₀ (mouse, IP): 1.297 g/kg

LD₅₀ (rabbit, oral): 5.340 g/kg

LD₅₀ (rabbit, skin): 0.2 g/kg